

Deepfake Detection via CLIP Feature Space

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Abstract

The exponential growth of synthetic images generated by advanced generative models has made distinguishing between real and computer-generated content increasingly difficult. To improve detection, recent methodologies have leveraged the robust feature space of Vision Transformers, particularly CLIP ViT-14. In this paper, we investigate the internal mechanics of the CLIP ViT-14 architecture to identify where and how synthetic artifacts are localized and detected across different layers and tokens. Our analysis reveals a specific, robust subset of spatial patches that exhibits highly stable behavior in exposing generative footprints. These findings offer new insights into the generalization limits of deepfake detectors across diverse generative architectures and families.

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1. Introduction

Generative adversarial networks (GANs), such as the **StyleGAN** family, and more recently **diffusion models** (such as Stable Diffusion), can now produce synthetic images that are hard to distinguish from real photographs. This raises real concerns for cybersecurity, media integrity, and it is one of the reasons why automatic deepfake detection is an active research area in digital forensics. Modern deep learning classifiers reach near-perfect accuracy when tested on the same dataset they were trained on (intra-dataset), but their performance drops significantly on images generated by architectures they have never seen (cross-architecture).

Recent work has started to leverage pre-trained vision-language models, in particular **CLIP** (Contrastive Language-Image Pre-training) with a Vision Transformer (ViT-14) backbone. Thanks to its large and general embedding space, CLIP tends to be more robust than task-specific classifiers at capturing both semantic and structural features.

In this work, we study the internal architecture of **CLIP ViT-14** to map where and how artificial artifacts are

detected. We analyze the network's responses at different depths (Transformer layers) and isolate the behavior of specific spatial regions of the image, dividing the patch grid into central (semantic) areas and corner (peripheral) areas.

The main contributions of this study can be summarized as follows:

- **Layer-wise Analysis:** We investigate the evolution of forgery discrimination across the 24 blocks of the CLIP Transformer, identifying the optimal levels where synthetic anomalies emerge most clearly.
- **Spatial Mapping (Center vs. Corners):** we demonstrate how generalization ability strongly depends on the analyzed region of the image.
- **Robust Patch Identification:** we identify a specific and robust subset of spatial patches that maintain stable and resilient behavior in artificial fingerprinting, laying the foundation for more lightweight, interpretable, and targeted sensing architectures.

This paper aims to support the study and development of a possible general DeepFake detector.

2. Related Work

2.1. Stable Diffusion vs GAN

These two generative models differ fundamentally in their synthesis processes:

- **GANs (Generative Adversarial Networks)** generate high-resolution images in a **one-shot** manner through adversarial learning, where a generator learns to up-sample noise to trick a discriminator.
- **Stable Diffusion** relies on a diffusion process. The model progressively adds Gaussian noise to a real image and is trained to iteratively reverse this process to reconstruct the original image.

2.2. The architecture of CLIP ViT-14

In this paper, **CLIP ViT-14** refers to OpenAI's **CLIP ViT-L/14**. Unlike standard Convolutional Neural Networks (CNNs), its Vision Transformer Large (ViT-L) encoder

processes images by dividing them into non-overlapping (14×14) pixel patches. Each patch is embedded as a token and passed through 24 Transformer layers. This global self-attention mechanism captures long-range dependencies, outputting a sequence of 1024-dimensional embeddings that encode rich visual and textual semantics.

2.3. Artifact Localization and Patch-Based Forensics

Early forgery detection focused on GAN-specific statistical anomalies. Chai et al. [1] pioneered patch-based classification, restricting the receptive field of CNNs to prove that local structural and textural artifacts are more robust synthetic indicators than global semantics. They also highlighted that generative footprints accumulate in specific regional clusters rather than being uniformly distributed. While patch-based CNNs excelled on GANs, their cross-architecture generalization degraded significantly with the advent of Stable Diffusion. Consequently, recent forensic frameworks have shifted towards multi-modal models like CLIP ViT-L/14 [2], whose large-scale contrastive pre-training captures both high-level semantic context and low-level pixel regularities. Building upon these foundations, our work extends the patch-level analysis of Chai et al. to the feature space of modern Vision Transformers. We systematically analyze CLIP ViT-L/14 layer-by-layer and patch-by-patch, mapping where the model’s embeddings reveal synthetic behavior and identifying which patches provide the strongest detection power.

3. Methodologies

The overall methodology is structured into three sequential phases: (i) layer-wise feature extraction via a frozen Vision-Language foundation model, (ii) targeted spatial patch isolation at specific depth levels, and (iii) independent linear classification to assess cross-architecture generalization.

3.1. Feature Extraction

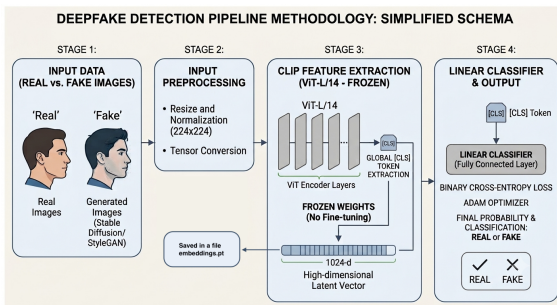


Figure 1. Pipeline of the Framework

We utilize the pre-trained CLIP ViT-L/14 as a frozen feature extractor to project images —resized to 224×224 pixels— into a robust latent space. To investigate at which level of abstraction synthetic artifacts are most prominent, we intercept intermediate token embeddings across specific Transformer layers:

- **StyleGAN1 and Stable Diffusion 1.4:** 12 layers (1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 21, and 23).
- **StyleGAN2, StyleGAN3, StyleGAN-XL, and Stable Diffusion 2.1:** 6 layers (3, 7, 11, 15, 19, and 23).

3.2. Spatial Token Extraction and Patch Selection

The ViT-L/14 backbone natively divides the input image into a 16×16 grid of non-overlapping 14×14 pixel patches, yielding a sequence of 256 spatial tokens plus a global Classification Token ([CLS]). To analyze the spatial distribution of generative footprints, we extract tokens corresponding to specific locations:

- **Peripheral Patches (Corners):** The four extreme grid corners (Top-Left, Top-Right, Bottom-Left, Bottom-Right) to capture boundary conditions and isolated padding artifacts.
- **Semantic Patches (Center):** The four central patches to isolate core semantic features, such as specific anatomical details in generated faces.
- **Classification Token ([CLS]):** Extracted as a baseline to summarize the overall structural and semantic information of the entire image grid.

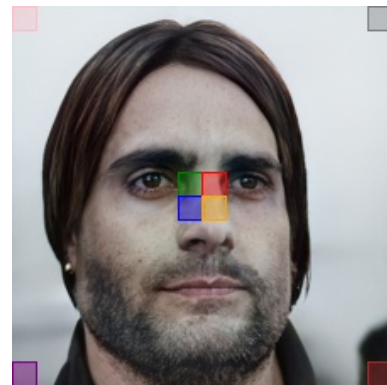


Figure 2. Visualization of the grid and the specific, individual spatial patches isolated by the framework.

3.3. Classification Paradigms

To evaluate both the linear separability and non-linear potential of the extracted embeddings, each spatial region is independently processed by three distinct classifiers:

- **Support Vector Classifier (svc):** Finds the optimal hyperplane separating real and synthetic features in the latent space.
- **Linear Classifier (linear):** A lightweight baseline to assess strict linear separability.
- **Multi-Layer Perceptron (MLP):** A fully-connected network designed to capture non-linear interactions and complex feature patterns.

Balanced class weighting and robust optimization are strictly applied across all models to mitigate potential dataset imbalances.

3.4. Evaluation Framework and Cross-Architecture Validation

To assess the robustness and generalizability of the detected synthetic artifacts, we evaluate the framework in two modes:

- **Intra-Family Mode (Standard Validation):** Models are trained and tested within the same generative domain (e.g., StyleGAN1 to StyleGAN3) to establish a baseline for internal semantic consistency and generational gap resilience.
- **Cross-Architecture Mode (Cross-Dataset Validation):** Models are trained on one specific architecture (e.g., GANs) and tested on a completely distinct target (e.g., Stable Diffusion) to evaluate structural generalization.

Performance is measured individually for each isolated patch at every extracted layer. By computing an independent **AUC** and **Accuracy** for each distinct patch-layer combination, this approach pinpoints exactly which spatial area and architectural depth yield the highest detection efficacy.

4. Experiments and Results

In this section, we detail the experimental evaluation of our proposed framework, discussing the findings and their implications for generalized deepfake detection.

4.1. Baseline Evaluation: Global Token Analysis Across Layers

In our initial experiment, we evaluated the detection capabilities of the global Classification Token ([CLS]) using one dataset of real images and two distinct datasets of synthetic images: StyleGAN1 and Stable Diffusion 1.4. Embeddings were extracted by performing a forward pass of each image through the frozen CLIP ViT-L/14 backbone, isolating the [CLS] token at twelve specific intermediate layers: 1, 3, 5, 7, 9, 11, 13, 15, 17, 19,

21, and 23. To determine the most effective classification paradigm, we trained independent classifiers for each layer and each generator. This required the training of 72 distinct models (12 layers $\times$ 2 generative datasets $\times$ 3 classifier types: Linear, SVM, MLP).

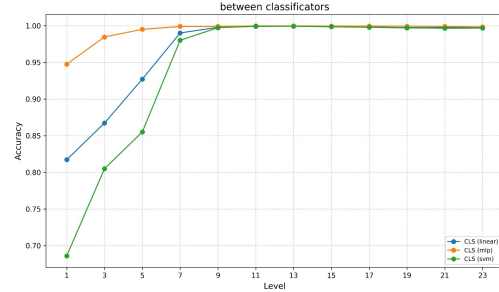


Figure 3. Intra-dataset Accuracy and AUC behavior of the linear classifier across extracted layers (Stable Diffusion 1.4).

4.1.1 Intra-Dataset Performance

We first established a performance baseline by evaluating the models on intra-dataset test samples (e.g., training and testing on the same generative architecture). The results utilizing the linear classifier are illustrated in Figures 3 and 5.

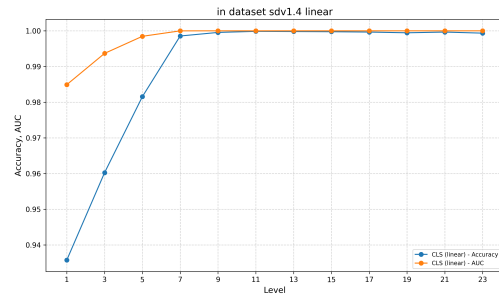


Figure 4. Intra-dataset Accuracy of the classifiers across extracted layers (StyleGAN1).

From here to the end, we will show the performance of the linear classifier.

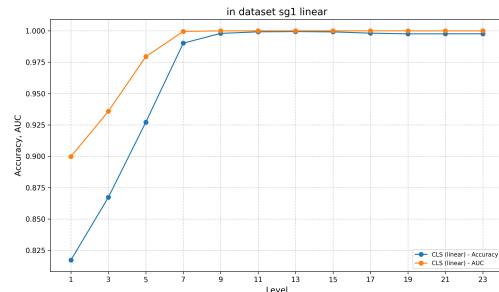


Figure 5. Intra-dataset Accuracy and AUC behavior of the linear classifier across extracted layers (StyleGAN1).

As observed in the plots, both the Area Under the Curve (AUC) and overall Accuracy exhibit a rapid upward trajectory in the early layers, taking place near a perfect score (1.0) consistently after the 7th layer. This confirms that the [CLS] token effectively captures the dataset-specific synthetic footprints at relatively shallow depths when tested within the same generative family.

4.1.2 Cross-Architecture Generalization

The near-perfect intra-dataset results naturally raise a critical question:

Does this structural representation generalize when tested against an unseen generative architecture? To investigate this, we performed cross-dataset evaluations, where classifiers trained on Stable Diffusion embeddings were tested on StyleGAN1 samples (Figure 6), and vice versa (Figure 7).

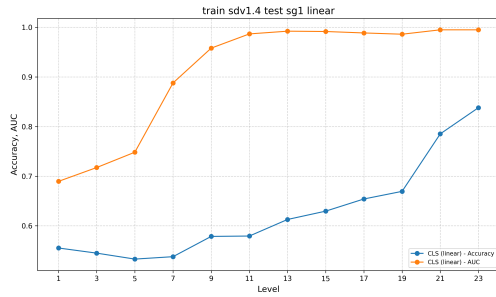


Figure 6. Cross-architecture evaluation: Linear classifier trained on Stable Diffusion 1.4 and tested on StyleGAN1.

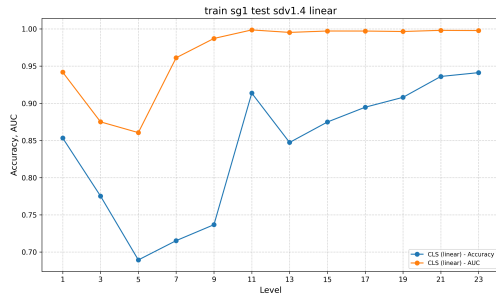


Figure 7. Cross-architecture evaluation: Linear classifier trained on StyleGAN1 and tested on Stable Diffusion 1.4.

Cross-architecture evaluation reveals severe performance degradation and instability compared to intra-dataset baselines. Although AUC eventually stabilizes in deeper layers, Accuracy metrics expose significant generalization flaws:

- **Stable Diffusion to StyleGAN1 (Figure 6):** Accuracy begins near random chance (≈ 0.55) in shallow layers and struggles to climb, exceeding 0.80 only at the very end of the network (layers 21 and 23).

- **StyleGAN1 to Stable Diffusion (Figure 7):** Performance exhibits a "V-shape" degradation, with accuracy dropping sharply below 0.70 around layer 5 before progressively recovering.

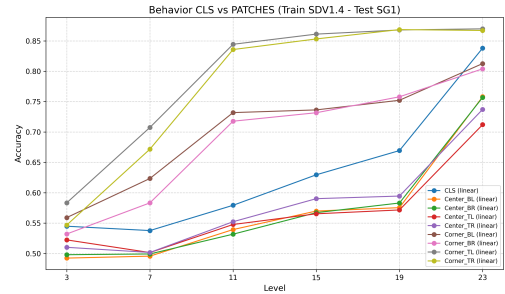
These results demonstrate that while the global [CLS] token eventually captures generalized features in the final Transformer blocks, its representation remains highly unstable and architecture-biased in the shallow and middle layers.

4.2. Spatial Token Dissection: The Superiority of Peripheral Patches

The cross-architecture limitations of the global [CLS] token suggest a fundamental flaw: its image-wide summary dilutes highly localized structural artifacts with dominant semantic features. To isolate the true generative footprints, we bypassed the global representation to analyze specific spatial tokens.

Following Section 3.2, we extracted eight spatial tokens from the 16×16 grid: four extreme peripheral patches (Corner_*) and four exact center patches (Center_*). To optimize computational efficiency while capturing architectural trends, we targeted a subset of six crucial Transformer layers (3, 7, 11, 15, 19, and 23).

The isolated performance of these spatial tokens, compared against the global [CLS] baseline, is detailed in Figures 8 and 9.



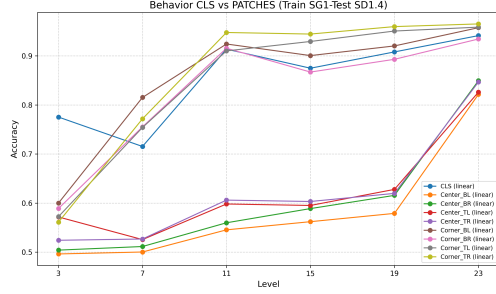


Figure 9. Cross-architecture Accuracy (Train: StyleGAN1 $\rightarrow$ Test: Stable Diffusion 1.4). Comparison between individual spatial patches (Corners and Centers) and the global [CLS] token.

From layer 11 onward, they clearly detach from the other tokens, surpassing 0.85 (Figure 8) and 0.90 (Figure 9) accuracy while avoiding the “V-shape” degradation of the [CLS] baseline.

These findings demonstrate that the most reliable indicators of synthetic generation are localized in the background corners since low levels. The global [CLS] token underperforms because it aggregates these robust peripheral signals with unreliable, domain-specific semantic features. Consequently, isolating peripheral tokens yields a significantly more robust detection signal.

4.3. Extensive Generalization Test: Evaluating Newer Generative Architectures

The previous experiment established that peripheral spatial patches (corners) outperform the global [CLS] token and central patches when transferring between StyleGAN1 and Stable Diffusion. To validate the universality of this structural footprint, we have to test these classifiers against entirely unseen, more advanced generative architectures, specifically StyleGAN3 and StyleGAN-XL. We took the classifiers previously trained the first two embeddings datasets, and tested them directly on StyleGAN3 and StyleGAN-XL samples.

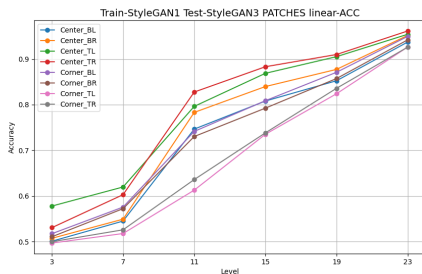


Figure 10. Cross-architecture evaluation (Train: StyleGAN1 $\rightarrow$ Test: StyleGAN3).

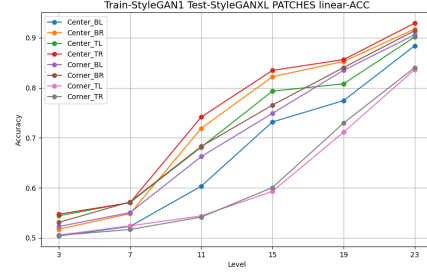


Figure 11. Cross-architecture evaluation (Train: StyleGAN1 $\rightarrow$ Test: StyleGAN-XL).

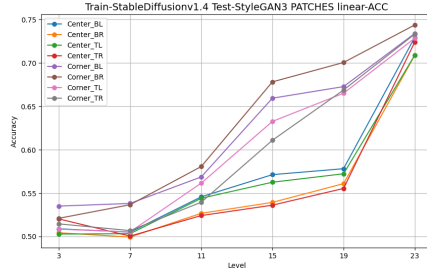


Figure 12. Cross-architecture evaluation (Train: Stable Diffusion 1.4 $\rightarrow$ Test: StyleGAN3).

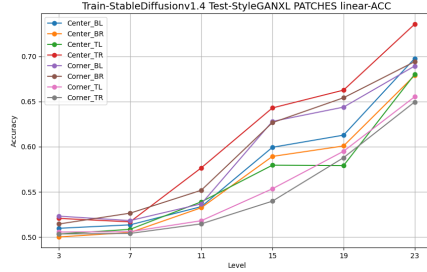


Figure 13. Cross-architecture evaluation (Train: Stable Diffusion 1.4 $\rightarrow$ Test: StyleGAN-XL).

4.3.1 Analysis of Semantic Domain vs. Structural Footprints

Extended tests reveal that the relative performance of central versus peripheral patches depends heavily on semantic domain overlap between the training and test sets.

- **Shared Domain (GAN to GAN):** When trained on StyleGAN1 and tested on StyleGAN3/XL (Figures 10 and 11), *central* semantic patches (Center.TR, Center.TL) actually outperform corners in intermediate layers (11-19). Because StyleGANs predominantly generate aligned faces, domain-specific semantic artifacts learned on SG1 transfer successfully to SG3 and SG-XL.

- **Cross-Domain (Diffusion vs. GAN):** When architectures diverge significantly, generalization becomes highly unstable and dataset-dependent. In Figure 11, peripheral corners maintain a slight advantage over central patches in the intermediate layers (11-19); however, overall accuracy remains remarkably low, failing to surpass 0.70 until the end of the network. Interestingly, Figure 12 shows a different dynamic: a specific central patch (Center_TR) actually outperforms the corners from layer 11 onward.

These results contradict the hypothesis of universal corner superiority in cross-domain scenarios. While corners capture robust generative artifacts in same-family models, testing a Diffusion-trained classifier on GANs reveals severe domain gaps where neither semantic centers nor structural corners can guarantee stable, high-accuracy generalization across all layers.

4.3.2 Asymmetric Generalization: Training on Advanced Generators (StyleGAN3 / XL)

We evaluated an inverted scenario: training the classification models on the embeddings StyleGAN3/XL and testing their performance backward on the older StyleGAN1, as well as across paradigms on Stable Diffusion 1.4. We will show the results with StyleGAN3 classifiers.

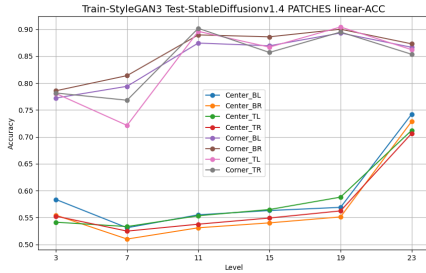


Figure 14. Cross-architecture evaluation (Train: StyleGAN3 → Test: Stable Diffusion 1.4).

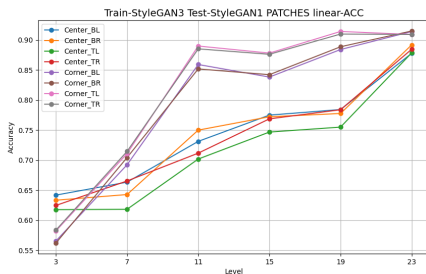


Figure 15. Cross-architecture evaluation (Train: StyleGAN3 → Test: StyleGAN1).

This test is relevant because StyleGAN3 and StyleGAN-XL are explicitly designed to suppress structural translation aliasing (the so-called "sticky pixels" artifact). We want to check whether a detector trained on these cleaner, more refined feature spaces can still identify the artifacts of older or completely different generative models.

The asymmetric evaluation gives two main results:

- **(StyleGAN3/XL → StyleGAN1):** When evaluated on StyleGAN1, the classifiers reach near-perfect accuracy at almost all layers. Because StyleGAN1 artifacts are a coarser version of the ones produced by StyleGAN3 so the refined StyleGAN3 features can still separate the more visible anomalies in StyleGAN1 images.
- **(StyleGAN3/XL → Stable Diffusion 1.4):** When transferring from these advanced GANs to diffusion models, as shown in Figure 14, all central patches (Center_*) fall to near-random accuracy ($\approx 0.50 - 0.60$) up to layer 19. The peripheral patches (Corner_*), on the other hand, remain robust from the start, separating from the central patches as early as layer 7 and staying above 0.95 accuracy through the deepest layers.

This shows that even when a Vision Transformer takes an image's advanced generator, its peripheral tokens (Corner_*) still capture low-level anomalies of the image synthesis process that are shared across both advanced GANs and diffusion models. This supports the idea that corner patches are a reliable anchor for cross-architecture deepfake detection.

4.4. Fourth Experiment: Cross-Generation Robustness and Asymmetric Peripheral Anchors

In this final experimental setup, we evaluate the ultimate generalization boundary of our patch-based approach. We select the linear classifiers trained exclusively on the feature space of the state-of-the-art StyleGAN3 architecture and perform a cross-dataset test against three distinct generative distributions embedded via the same CLIP model:

- **StyleGAN2:** To observe backward inheritance and architectural consistency within the same generative family.
- **StyleGAN-XL:** To observe upscaled, modern GAN scaling effects.
- **Stable Diffusion 2.1:** To evaluate transferability to the latest iteration of latent diffusion models.

The layer-wise accuracy patterns across these configurations are illustrated in Figures 16, 17, and 18.

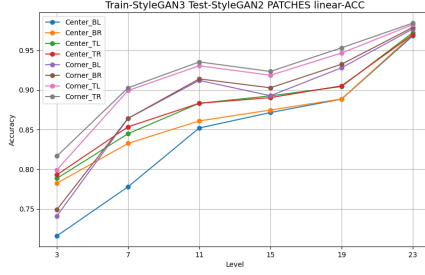


Figure 16. Cross-dataset validation within the GAN family (Train: StyleGAN3 → Test: StyleGAN2).

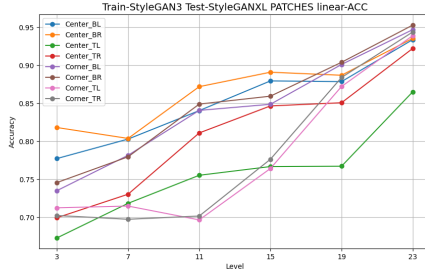


Figure 17. Cross-dataset validation on upscaled architectures (Train: StyleGAN3 → Test: StyleGAN-XL).

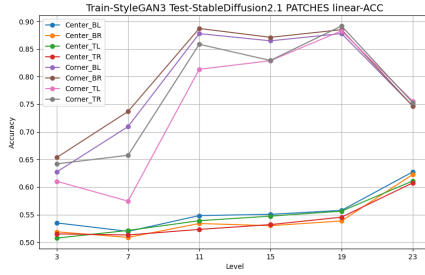


Figure 18. Cross-paradigm evaluation on advanced diffusion spaces (Train: StyleGAN3 → Test: Stable Diffusion 2.1).

4.4.1 Discovery of Lower-Peripheral Asymmetric Robustness

These final validation charts reveal a consistent structural weakness across generative paradigms.

During intra-family testing on StyleGAN2 (Figure 16) and StyleGAN-XL (Figure 17), both central and peripheral tokens perform equally well, reaching near-perfect detection ($\approx 0.95 - 0.98$) at layer 23.

Conversely, cross-paradigm evaluation against Stable Diffusion 2.1 (Figure 18) exposes a stark contrast: central tokens (*Center_**) collapse to near-random accuracy ($\approx 0.50 - 0.55$) in early-to-intermediate layers, while peripheral tokens remain highly robust.

The lower peripheral patches—**Corner_BL** and

Corner_BR—significantly outperform the top corners, achieving > 0.78 accuracy by layer 3 and peaking at ≈ 0.89 in intermediate layers (11–19).

This suggests that lower corners capture the purest structural footprints of the generative process, whereas top corners are often polluted by variable semantic content (like hair). Consequently, lower peripheral patches act as the most reliable anchors for cross-architecture deepfake detection, generalizing effectively from advanced GANs to latent diffusion models.

4.4.2 ROC Curve Validation: Threshold-Independent Confirmation

To confirm the robustness of lower peripheral tokens independently of fixed classification thresholds, we evaluate the feature space separability using ROC curves and Area Under the Curve (AUC) metrics. We compare the diagnostic performance of the *Corner_BL* and *Center_BL* patches at two critical depths: Layer 7 (Figures 19 and 20) and Layer 11 (Figures 21 and 22).

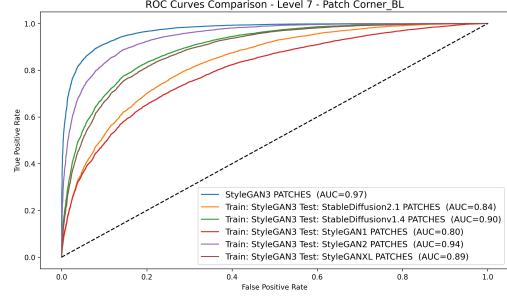


Figure 19. ROC curves at Layer 7 for the individual *Corner_BL* spatial token across unseen testing distributions.

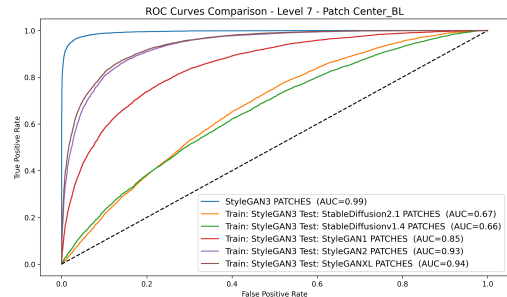


Figure 20. ROC curves at Layer 7 for the individual *Center_BL* spatial token across unseen testing distributions.

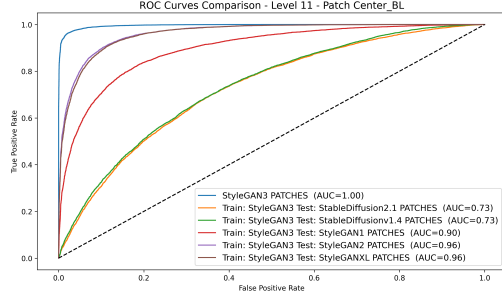


Figure 21. ROC curves at Layer 11 for the individual `Center_BL` spatial token across unseen testing distributions.

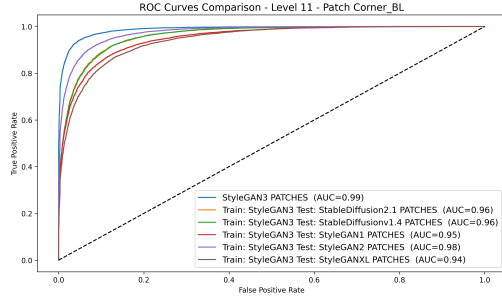


Figure 22. ROC curves at Layer 11 for the individual `Corner_BL` spatial token across unseen testing distributions.

The threshold-independent ROC analysis strongly supports our previous findings:

- **Layer 7:** The `Center_BL` token (Figure 20) successfully isolates intra-family models (StyleGAN2) but collapses to random guessing ($AUC \approx 0.50$) for cross-paradigm targets (Stable Diffusion 1.4 and 2.1). Conversely, the `Corner_BL` token (Figure 19) already lifts the Stable Diffusion curves well above the baseline.
- **Layer 11:** The generalization gap widens. The central patch (`Center_BL`, Figure 21) remains entangled by semantic overfitting, failing to separate latent diffusion datasets. Meanwhile, the peripheral patch (`Corner_BL`, Figure 22) achieves high AUC scores across *all* evaluated datasets, with curves approaching the ideal top-left corner for both GAN and Diffusion families.

This analysis validates our core hypothesis: central patches track architecture-dependent semantic features, whereas lower corners capture universal structural anomalies that remain clearly separable regardless of the underlying generative paradigm.

5. Conclusion and Future Work

We introduced a token-level dissection framework to investigate deepfake detection generalization across distinct

generative families (GANs vs. Latent Diffusion). By isolating individual spatial tokens from the intermediate layers of a frozen CLIP ViT-L/14 model, our experiments yield three main conclusions:

1. **Central tokens overfit semantics:** Central patches (`Center_*`) are heavily tied to the training domain. They excel in intra-domain tasks but fail to generalize across paradigms that synthesize distinct subjects (e.g., GANs to Stable Diffusion).
2. **Global token dilution:** The global `[CLS]` token exhibits domain bias and instability in early-to-intermediate blocks. This occurs because its global representation aggregates robust structural signals with the unreliable, overfitting semantic features of the central regions.
3. **Lower corners are the most reliable anchors:** Lower peripheral patches (`Corner_BL`, `Corner_BR`) capture the most robust and linearly separable synthetic footprints. Threshold-independent ROC analyses confirm high cross-architecture accuracy ($\approx 0.85 - 0.95$) against unseen models (StyleGAN3, StyleGAN-XL, and Stable Diffusion 2.1). This demonstrates that low-level geometric artifacts (e.g., padding and grid-anchoring) systematically accumulate at image boundaries regardless of the underlying generation process.

Future work will explore designing lightweight, semantics-free detectors by geometrically combining these robust peripheral tokens. We also plan to extend this analysis to other vision-language foundation models to further verify the universality of boundary-localized footprints.

References

- [1] Chai, L., Bau, D., Lim, S. N., & Isola, P. (2020). *What makes fake images detectable? Understanding properties that generalize*. In *European Conference on Computer Vision (ECCV)* (pp. 103-120). Springer, Cham.
- [2] Radford, A., Kim, J. W., Hallacy, C., Ramesh, A., Goh, G., Agarwal, S., Sastry, G., Askell, A., Mishkin, P., Clark, J., Krueger, G., & Sutskever, I. (2021). *Learning transferable visual models from natural language supervision*. In *International Conference on Machine Learning (ICML)* (pp. 8748-8763). PMLR.